

5 Interesting concepts I learnt from the lectures

1. **The Tiers of Computer Vision:** I loved how the field is structured into low-level (pixel manipulation and edge detection), mid-level (connecting images to 3D space like panorama stitching), and high-level vision (semantic understanding like object detection and image classification).
2. **The Biology of Human Vision:** It was super interesting to see how our eyes work compared to cameras. The lectures explain how we use rods for seeing in the dark and cones for color, and how our brains process all this information behind the scenes.
3. **Light and Color Spaces:** I really enjoyed the explanation of metamers and human color perception. It made perfect sense how we use RGB representations, and why switching to Hue, Saturation, and Value (HSV) allows for much intuitive image manipulation.
4. **Digital Image Representation:** Seeing the breakdown of how computers actually store images was eye-opening. I liked learning about the pinhole camera model, Bayer patterns for color filtering, and how images translate into 3D coordinate tensors in memory.
5. **Resizing and Interpolation:** I liked the breakdown of how we resize photos. If you just stretch a tiny image, it gets terribly blocky. The computer actually has to guess the missing pieces by looking at the nearby colors to make the bigger picture smooth.

Types of Healthcare Reports

1. Diagnostic/Laboratory Reports

Documents detailing results from medical tests and imaging used to identify, diagnose, and monitor various health conditions.

- i. **Blood test:** Checks vital markers like sugar levels, TSH, WBC, and RBC counts to assess your overall health status.
- ii. **CT scan:** A painless, non-invasive test using X-rays from multiple angles to create detailed, processed images of the body.
- iii. **Histopathology:** Examines tissue slides under a microscope to study the microscopic manifestations and progression of specific diseases.
- iv. **X-rays:** Uses quick electromagnetic radiation to observe and diagnose internal injuries, fractures, and cracks within the skeletal system.
- v. **MRI:** A non-invasive test using strong magnets and radio waves to generate detailed images of internal organs and tissues.

2. Clinical Reports

Records of patient visits, ongoing medical evaluations, and daily treatment plans created by healthcare providers.

i. Consultation reports: A specialist's expert opinion and evaluation regarding a patient's specific condition, requested by the primary care doctor.

ii. SOAP notes: A structured method used by providers to document a patient's Subjective complaints, Objective findings, Assessment, and Plan.

3. Procedural/Hospitalisation Reports

Detailed documents recording medical procedures, surgeries, and significant events that happen during a patient's hospital stay.

i. Operation report: Details a surgical procedure, including the techniques used, surgical findings, and the patient's immediate post-operative condition.

ii. Autopsy report: A specialized medical examination of a deceased person to discover the exact cause and manner of death.

Anatomy of a Healthcare Report

Healthcare reports, specifically in radiology, follow a highly standardized structure. This ensures that the physician who ordered the test gets exactly the information they need without any confusion. A typical report is broken down into four main parts:

- **Clinical Indication:** The "why." This is a brief note explaining the patient's symptoms or the reason the test was ordered (e.g., "chronic cough").
- **Technique:** The "how." This details the specific machinery settings used, the angles taken, and whether any contrast dye was given to the patient.
- **Findings:** The objective observations. The specialist goes through every organ, bone, or tissue visible in the scan and describes what they see, whether it looks normal or abnormal.
- **Impression:** The bottom line. This is the conclusion of the report, offering a definitive diagnosis or a short list of possible conditions, along with recommendations for next steps.

Medical Imaging Reports:

1. **X-Rays:** Uses a quick flash of electromagnetic radiation that passes through soft tissue but is absorbed by dense materials like bone. This creates a clear, two-dimensional image ideal for diagnosing fractures or lung issues.

2. **CT Scan (Computed Tomography):** Takes multiple X-ray images from different angles around the patient. A computer processes these inputs to generate detailed, cross-sectional images of organs, bones, and blood vessels.
3. **MRI (Magnetic Resonance Imaging):** Utilizes a powerful magnetic field and radio waves instead of radiation. This interacts with protons in the body to produce highly detailed images, making it the standard for examining soft tissues such as the brain, spinal cord, and ligaments.
4. **Ultrasound:** Emits high-frequency sound waves into the body using a transducer. The machine records the echoes bouncing off internal structures to generate real-time images, making it safe for monitoring fetal development or blood flow without radiation.